

Andrew Nash

Staff Autonomous Driving Engineer | Simulation | Localization | Planning | Perception

@ anash1324@gmail.com 📍 Ottawa, Canada | open to relocation in LinkedIn GitHub Portfolio

Summary. Self-driving tech lead and core author of a Transport Canada-authorized SAE Level 4 autonomy stack: simulation, sensor fusion and localization, motion planning, and machine learning perception.

Experience

Aurrigo

Staff ADS Engineer

📅 Oct 2025 - Present 📍 Ottawa, Canada

- **Led autonomy for Canada’s first medium-speed autonomous shuttle:** team of 6; core author and code owner of two \$1M+ deployments [press].
- **Built an airport digital twin 50x faster than CARLA:** aircraft turnaround (vehicles, people, baggage); CUDA sensors, LiDAR 6x faster than NVIDIA’s.
- **Authored production localization at 5 cm lateral accuracy:** real-time GNSS, LiDAR and IMU sensor fusion; live at 5 sites on 4 vehicle platforms.
- **Authored a CUDA LiDAR perception pipeline:** occupancy mapping, SLAM and obstacle clustering in real time; 5 ms per scan on 0.1 CPU core.
- **Shipped camera-LiDAR 3D object detection (83.4 mAP, 20 Hz):** TensorRT inference on the vehicle with an ISO 26262 redundant safety channel.
- **Architected the fleet dashboard:** real-time WebGPU rendering of HD maps, live sensors, plans and vehicle views at a steady 60 FPS.

ROS C++ Python CUDA PyTorch TensorRT GTSAM Linux

Aurrigo

ADS Team Lead, North America

📅 Oct 2023 - Oct 2025 📍 Ottawa, Canada

- **Solo-authored LiDAR SLAM and mapping for 50+ km/h autonomy:** C++ GTSAM factor graph built on GLIM, fusing LiDAR, IMU and GPS.
- **Authored real-time C++ motion planning and control:** 400+ candidate trajectories at 50 Hz, selecting the smoothest safe path.
- **Built Aurrigo’s first AV simulator (Unreal Engine):** 1,000s of safety cases.

ROS C++ Python GTSAM SLAM Unreal Engine CI/CD

Internships

Nasdaq (Verafin) | Machine Learning Engineer

May - Aug 2021

Lifted gradient-boosting fraud-detection model precision by 6 pp at fixed recall.

Python Pandas Scikit-Learn XGBoost LightGBM AWS

Bank of Canada | Data Science Developer

Sept 2020 - Apr 2021

Built anomaly-detection models automating economist data-review workflows.

Python Pandas Scikit-Learn XGBoost JavaScript Tableau

IBM | NLP Developer

Jan - Apr 2020

Improved enterprise semantic matching by 8% via weighted word embeddings.

Python NLP Keras NLTK Java JavaScript Kubernetes Cognos

Royal Bank of Canada | Full Stack Data Scientist

May - Aug 2019

Built full-stack semantic search prototype with NLP backend and web frontend.

Python NLP BERT Flask React JavaScript Docker

BlackBerry QNX | NLP Developer

Sept - Dec 2018

Applied NLP classifiers to automate software-license tagging across QNX repos.

Python NLP Keras FastText QNX Docker MongoDB

Education

MSc Computer Science (Thesis)

Memorial University Of Newfoundland

📅 2021 - 2023 📍 NL, Canada

4.0 GPA. Thesis: Herds Eye View (AIIDE-2023).

BEng Computer Engineering (Co-Op)

Memorial University Of Newfoundland

📅 2016 - 2021 📍 NL, Canada

Capstone: SR AUV (best 2021 capstone).

Publications

Herds Eye View

Accepted to AIIDE-2023

🔗 arXiv 🔗 Blog Post

Unity RL Transformer PyTorch

Projects

Sage | Real-time D&D AI Co-pilot

🔗 Blog Post

- Live voice agent: per-player speech-to-text, LLM, 53 MCP tools, hybrid RAG.
- Distilled a 66M DistilBERT gate (70 ms on CPU); fine-tuned Qwen LoRAs up to 27B.

PyTorch LLM LoRA RAG MCP

IGVC | Autonomous Vehicle Competition

🔗 Blog Post

- Trained custom transformer BEV prediction models from Unreal Engine data.
- Design placed 2nd internationally 2023.

ROS Unreal Engine C++ PyTorch

SR AUV | Autonomous Underwater Vehicle

🔗 Blog Post

- Real-world tested end-to-end flying autonomously using PPO RL.
- Awarded best 2021 Capstone Project.

Unity C# TensorFlow TPU RL

SpaceX Hyperloop | Pod Competition

- Top 2.6% worldwide, 3rd in North America (2019); built full-stack pod telemetry.

C++ Flask JS Protobuf CAN